Identical Asymptotics with Different Finite Iterates: A Proof of Conjecture 00000006090

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Abstract

For any prescribed finite iteration, two gradient-descent step schedules on the same quadratic objective, with the same initial value, can produce different errors and then rejoin exactly one step later. Both sequences converge to the same minimizer and have asymptotic successive-error ratio 1/2. In fact, their entire tails coincide. The objective derivative, actual iterative algorithms, explicit trajectories, exact error gap, convergence, and quantified existence claim are verified in Lean.

Reading of the source. Both versions describe the desired distinction as different "finite-scale behavior" or "pre-asymptotics," despite the same limit and asymptotic convergence rate. We realize this using actual errors of iterative algorithms at finite iteration indices. The source does not define a separate operator or numerical invariant called the finite-scale spectrum, and none is introduced here. We make the convergence rate precise by the standard successive-error ratio and also prove the stronger property that the trajectories eventually coincide exactly. Thus the distinction cannot depend on choosing a weak interpretation of equal asymptotic rates.

Theorem 1. *Let $f(x) = x^2/2$ on $\mathbb{R}$ and start gradient descent at $x_0 = y_0 = 1$. For every integer $m \geq 0$, there exist two step-size schedules $a_n, b_n \in (0, 1)$ such that*

$$x_{n+1} = x_n - a_n f'(x_n), \qquad y_{n+1} = y_n - b_n f'(y_n)$$

and the following hold:

1. *$x_n = y_n$ for $0 \leq n \leq m$;*
2. *at the prescribed next iteration,*

$$|y_{m+1}| - |x_{m+1}| = 2^{-(m+2)} > 0;$$

3. *$x_n = y_n$ for every $n \geq m + 2$;*
4. *both sequences converge to the unique minimizer zero and*

$$\lim_{n\to\infty} \frac{|x_{n+1}|}{|x_n|} = \lim_{n\to\infty} \frac{|y_{n+1}|}{|y_n|} = \frac{1}{2}.$$

Consequently, the algorithms have the same asymptotic convergence rate and limit, but different finite-scale behavior.

Proof. The objective is nonnegative, vanishes exactly at zero, and has derivative $f'(x) = x$. Define

$$a_n = \frac{1}{2}, \qquad b_n = \begin{cases} \frac{1}{4}, & n = m, \\ \frac{2}{3}, & n = m + 1, \\ \frac{1}{2}, & \text{otherwise.} \end{cases}$$

Each step lies strictly between zero and one. Gradient descent becomes $x_{n+1} = (1 - a_n)x_n$ and $y_{n+1} = (1 - b_n)y_n$. Thus

$$x_n = 2^{-n}, \qquad y_n = \begin{cases} \frac{3}{2}\,2^{-n}, & n = m+1, \\ 2^{-n}, & n \neq m+1. \end{cases} \tag{1}$$

To check the second formula directly, the sequences follow the same updates up to index m, where both equal 2^{-m}. The changed step $1/4$ then gives

$$y_{m+1} = \frac{3}{4}\,2^{-m} = \frac{3}{2}\,2^{-(m+1)}.$$

The next changed step compensates exactly:

$$y_{m+2} = \left(1 - \frac{2}{3}\right) y_{m+1} = \frac{1}{3} \cdot \frac{3}{4}\,2^{-m} = 2^{-(m+2)} = x_{m+2}.$$

All later steps again equal $1/2$, so the trajectories remain identical. These calculations prove (1); alternatively its substitution in the recurrence proves it by induction.

All displayed iterates are positive. Therefore at $m+1$ the exact error difference is

$$|y_{m+1}| - |x_{m+1}| = \left(\frac{3}{2} - 1\right) 2^{-(m+1)} = 2^{-(m+2)}.$$

The geometric sequence 2^{-n} tends to zero. By eventual equality, so does y_n. The successive-error ratio of x_n equals $1/2$ at every index. For $n \geq m+2$, both y_n and y_{n+1} equal the corresponding x iterates, so its ratio is also exactly $1/2$. This proves all assertions. Since m is arbitrary, the differing finite iteration can be placed as late as desired, while the tails and therefore the asymptotic behavior still coincide exactly. □

Formal verification and scope

The file `lean/Main.lean` uses Lean 4.19.0 with Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

The objective is a real function and its derivative is verified using Mathlib's `HasDerivAt`. The two algorithms are actual recursively defined gradient-descent iterates using Mathlib's `deriv`, not numerical sequences named as algorithms without proving their update rule. The same objective and initial value are used for every m.

Lean proves the explicit trajectories and the step bounds $(0, 1)$. It proves equality of the prefixes and tails, the exact error gap, and the finite error discrepancy. Convergence to zero is a filter limit. The predicate `HasRate` gives the limit of the ratio of successive absolute errors. It is proved for both sequences. Their eventual equality is also formalized directly.

The final theorem is

`Conjecture6090.conjecture_6090_true`.

It proves `SeparationClaim`: there exists a single differentiable objective with unique zero minimum, such that for every m there exist two positive step schedules and their gradient-descent trajectories with the stated limit, rate, finite error discrepancy, common prefix, and identical tail. This is stronger than the source's existential claim. All mathematical ingredients needed for the theorem are formalized; no unspecified spectral object is substituted for the actual finite error behavior.

The audited main, derivative, trajectory, gap, and rate theorems use only `propext`, `Classical.choice`, and `Quot.sound`. This submission was developed by a delegated AI agent with a self-review; the coordinating agent reviews it separately. No external independent review is claimed.

Source and reproduction

The exact bilingual source is included as `SOURCE.md`, from upstream commit

`b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`.

The source hash and retrieval record are in

`verification/SOURCE_PROVENANCE.md`.

The proof is elementary and self-contained. With the pinned dependencies available, run in the `lean` directory:

```
lake env lean -DwarningAsError=true Main.lean
```

and `lake build` for a fresh build. Compile `main.tex` with Tectonic or a compatible LaTeX engine.